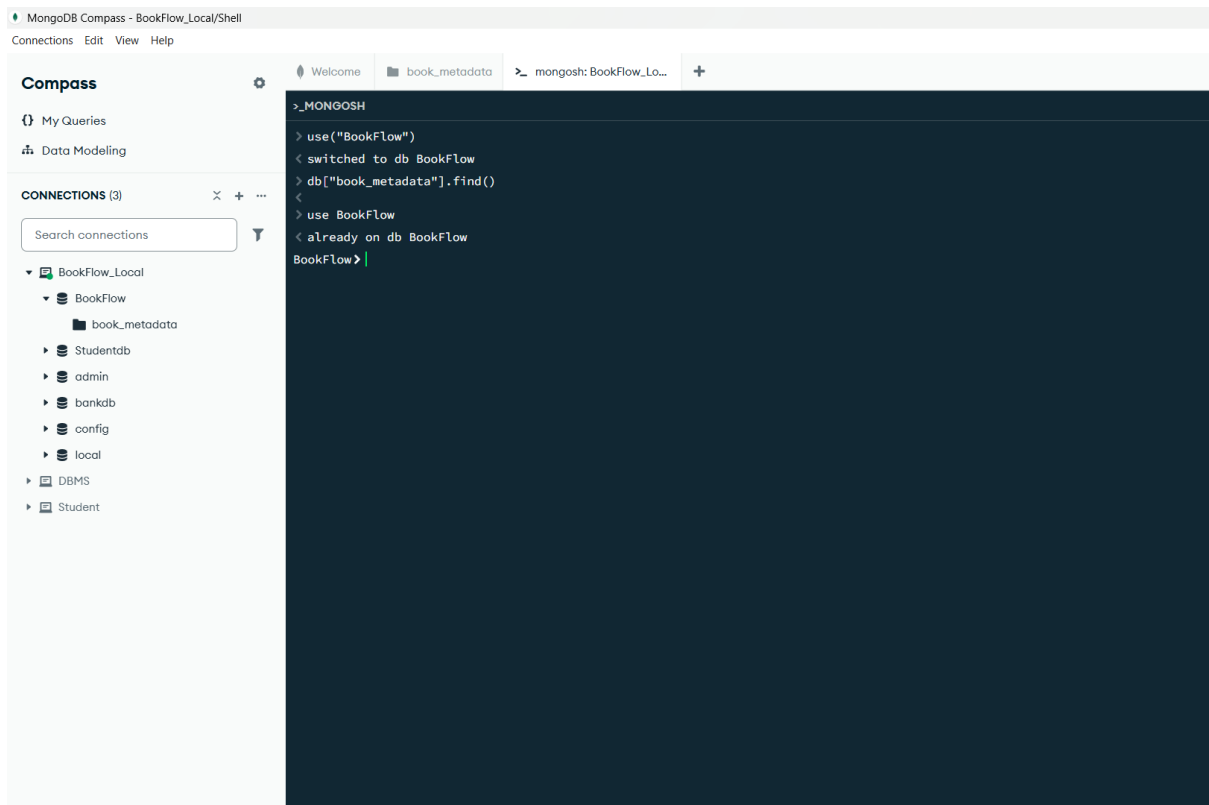


# DBMS WEEK-3

2520090016

P.Rahul

## Create Database



## Create book\_metadata Collection and Insert Data

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

### Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

book\_metadata

Studentdb

admin

bankdb

config

local

DBMS

Student

Welcomebook\_metadata\_mongosh: BookFlow\_25...

> MONGOSH

> use("BookFlow")  
< switched to db BookFlow  
> db["book\_metadata"].find()  
<  
> use BookFlow  
< already on db BookFlow  
> db.book\_metadata.insertMany([  
 {  
 title: "The Silent Algorithm",  
 author: "R. Menon",  
 genre: "Technology",  
 publishedYear: 2022,  
 formats: ["Hardcover", "Digital", "Audiobook"],  
 specs: {  
 pages: 340,  
 fileSize\_MB: 4.2,  
 drm: true,  
 narrator: "K. Rao",  
 runtime\_min: 610  
 },  
 avgRating: 4.6,  
 reviews: [  
 {  
 member\_id: "M1001",  
 rating: 5,  
 comment: "Excellent deep dive into digital systems design.",  
 date: ISODate("2023-01-15")  
 },  
 {  
 member\_id: "M1042",  
 rating: 4,  
 comment: "Good but the Digital edition had formatting issues.",  
 date: ISODate("2023-02-20")  
 }  
 ]  
 },  
 {  
 title: "Ocean of Code",  
 author: "A. Fernandes",  
 genre: "Technology",  
 publishedYear: 2021,  
 formats: ["Digital", "Paperback"],

## Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

book\_metadata

Studentab

admin

bankab

config

local

DBMS

Student

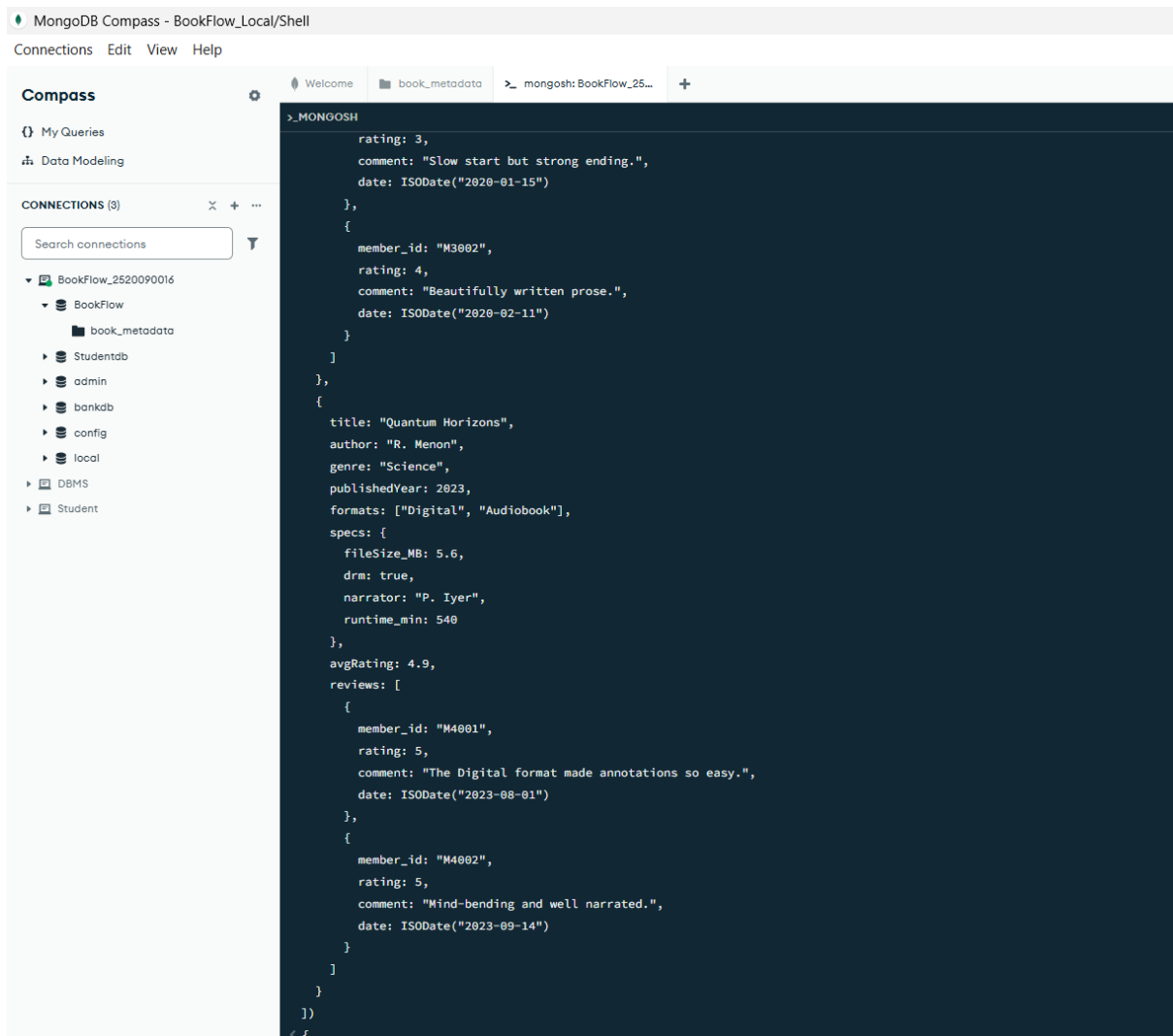
Welcome

book\_metadata

mongosh: BookFlow\_25...

&gt; MONGOSH

```
author: "A. Fernandes",
genre: "Technology",
publishedYear: 2021,
formats: ["Digital", "Paperback"],
specs: {
  pages: 210,
  fileSize_MB: 2.1,
  drm: false
},
avgRating: 4.2,
reviews: [
  {
    member_id: "M2001",
    rating: 4,
    comment: "Solid Digital read, clean layout.",
    date: ISODate("2022-05-10")
  },
  {
    member_id: "M2002",
    rating: 5,
    comment: "Best programming book this year.",
    date: ISODate("2022-06-01")
  }
]
},
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,
  formats: ["Hardcover", "Paperback"],
  specs: {
    pages: 410,
    binding: "Sewn"
  },
  avgRating: 3.8,
  reviews: [
    {
      member_id: "M3001",
      rating: 3,
      comment: "Slow start but strong ending.",
```



For Checking inserted documents: **db.book\_metadata.find().pretty()**

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

- BookFlow\_2520090016
  - BookFlow
    - book\_metadata
  - Studentab
    - admin
    - bankdb
    - config
    - local
  - DBMS
  - Student

Welcome

book\_metadata

mongosh: BookFlow\_25...

+

> MONGOSH

```
acknowledged: true,
insertedIds: {
  '0': ObjectId('6a8d2225a10553b67ccc28a3'),
  '1': ObjectId('6a8d2225a10553b67ccc28a4'),
  '2': ObjectId('6a8d2225a10553b67ccc28a5'),
  '3': ObjectId('6a8d2225a10553b67ccc28a6')
}
}
> db.book_metadata.find().pretty()
< [
  {
    _id: ObjectId('6a8d2225a10553b67ccc28a3'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [
      'Hardcover',
      'Digital',
      'Audiobook'
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: 2023-01-15T00:00:00.000Z
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: 2023-02-20T00:00:00.000Z
      }
    ]
  }
]
```

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

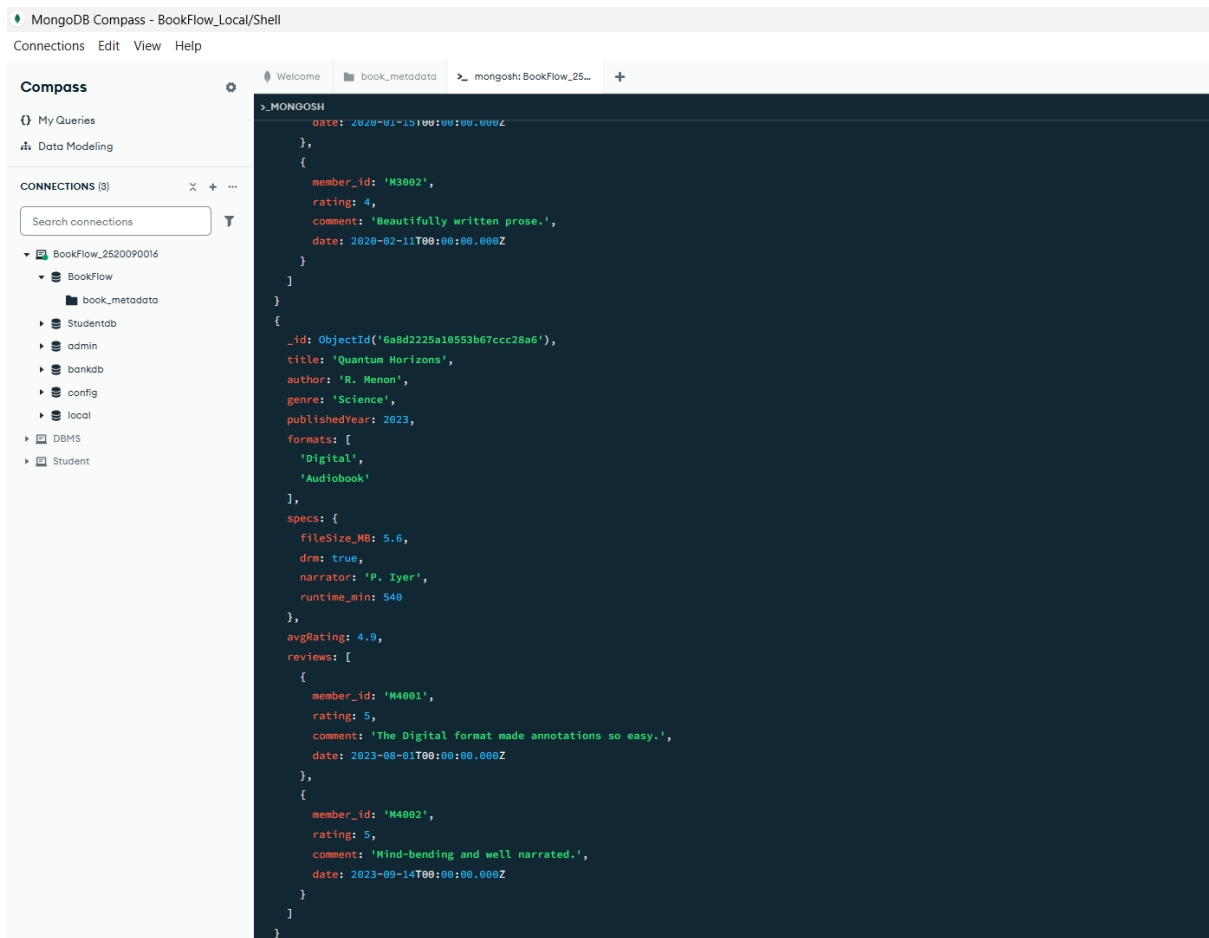
Search connections

- BookFlow\_2520090016
  - BookFlow
    - book\_metadata
    - Studenttab
    - admin
    - bankdb
    - config
    - local
  - DBMS
  - Student

> MONGOSH

```
{
  "_id": ObjectId('6a8d2225a18553b67ccc28a4'),
  "title": 'Ocean of Code',
  "author": 'A. Fernandes',
  "genre": 'Technology',
  "publishedYear": 2021,
  "formats": [
    'Digital',
    'Paperback'
  ],
  "specs": {
    "pages": 210,
    "fileSize_MB": 2.1,
    "drm": false
  },
  "avgRating": 4.2,
  "reviews": [
    {
      "member_id": 'M2001',
      "rating": 4,
      "comment": 'Solid Digital read, clean layout.',
      "date": 2022-05-10T00:00:00.000Z
    },
    {
      "member_id": 'M2002',
      "rating": 5,
      "comment": 'Best programming book this year.',
      "date": 2022-06-01T00:00:00.000Z
    }
  ]
}

{
  "_id": ObjectId('6a8d2225a18553b67ccc28a5'),
  "title": 'Whispers of Yesterday',
  "author": 'S. Kapoor',
  "genre": 'Fiction',
  "publishedYear": 2019,
  "formats": [
    'Hardcover',
    'Paperback'
  ],
  "specs": {
```



## Task 1 – Document Modeling

Display only title, author, formats and reviews:

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

book\_metadata

Studentdb

admin

bankdb

config

local

DBMS

Student

book\_metadata

mongo: BookFlow\_25...

```
>_MONGOSH
> db.book_metadata.find(
  {},
  {
    _id: 0,
    title: 1,
    author: 1,
    formats: 1,
    reviews: 1
  }
).pretty()
< {
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
{
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  formats: [
    'Digital',
    'Paperback'
  ],
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: 2022-06-01T00:00:00.000Z
    }
  ]
}
{
  title: 'Whispers of Yesterday',
  author: 'S. Kapoor',
  formats: [
    'Hardcover',
    'Paperback'
  ],
  reviews: [
    {
      member_id: 'M3001',
      rating: 3,
      comment: 'Slow start but strong ending.',
      date: 2020-01-15T00:00:00.000Z
    },
    {
      member_id: 'M3002',
      rating: 4,
      comment: 'Beautifully written prose.',
      date: 2020-02-11T00:00:00.000Z
    }
  ]
}
{
  title: 'Quantum Horizons',
  author: 'R. Menon',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
```

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

book\_metadata

Studentdb

admin

bankdb

config

local

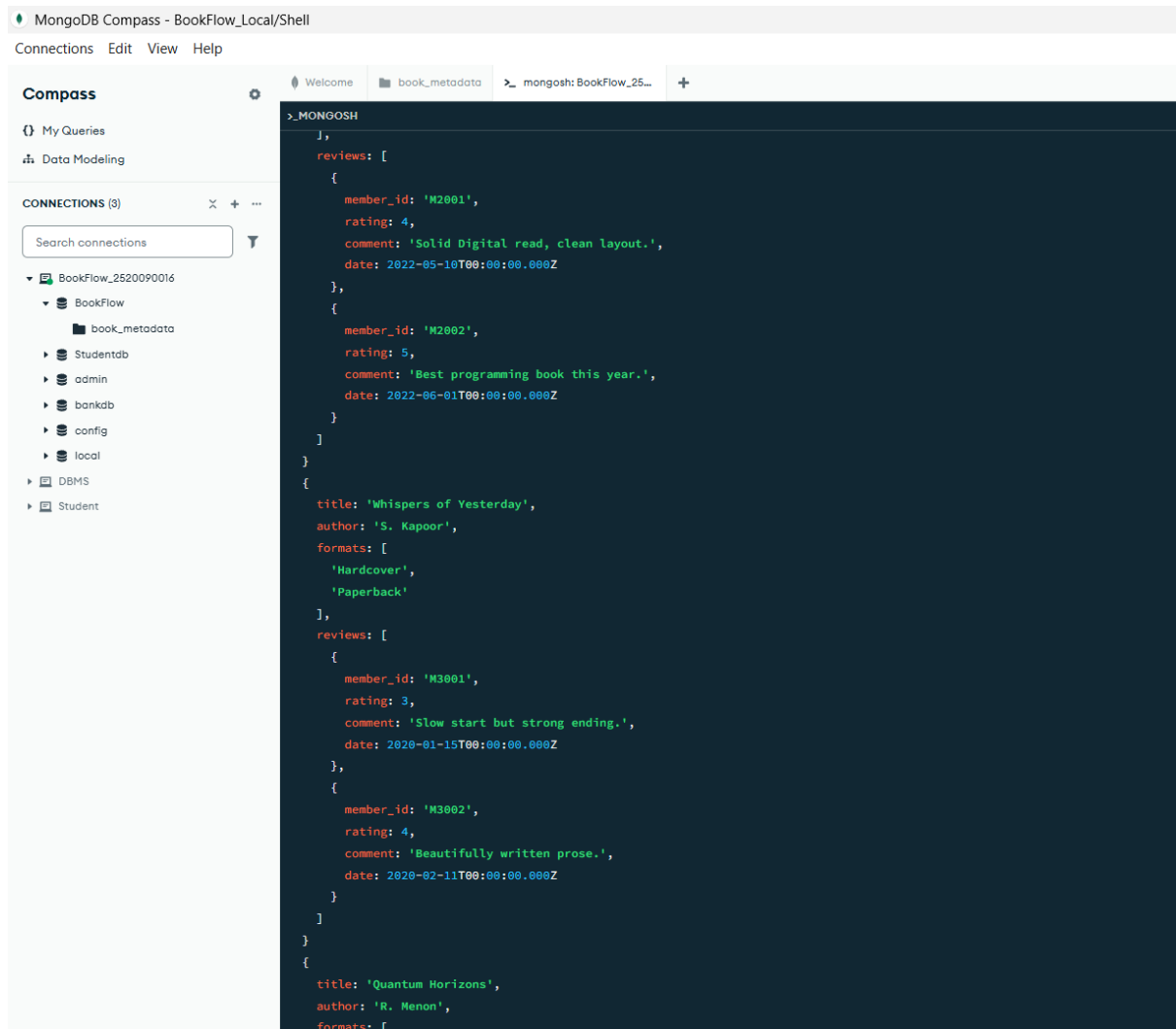
DBMS

Student

book\_metadata

mongo: BookFlow\_25...

```
>_MONGOSH
{
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
{
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  formats: [
    'Digital',
    'Paperback'
  ],
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: 2022-06-01T00:00:00.000Z
    }
  ]
}
{
  title: 'Whispers of Yesterday',
  author: 'S. Kapoor',
  formats: [
    'Hardcover',
    'Paperback'
  ],
  reviews: [
    {
      member_id: 'M3001',
      rating: 3,
      comment: 'Slow start but strong ending.',
      date: 2020-01-15T00:00:00.000Z
    },
    {
      member_id: 'M3002',
      rating: 4,
      comment: 'Beautifully written prose.',
      date: 2020-02-11T00:00:00.000Z
    }
  ]
}
{
  title: 'Quantum Horizons',
  author: 'R. Menon',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
```



Display books with their specifications:

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

### Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

book\_metadata

Studentab

admin

bankab

config

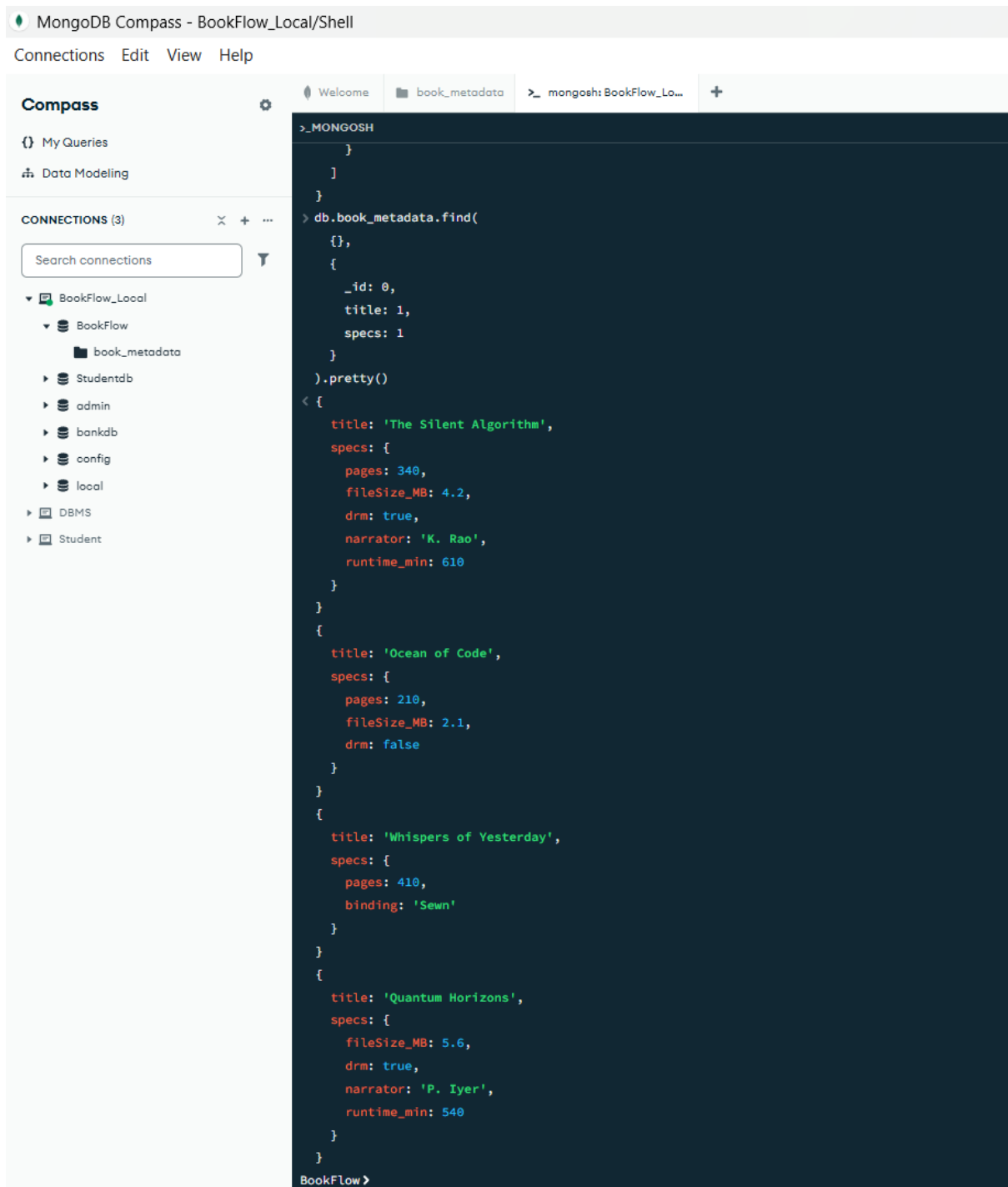
local

DBMS

Student

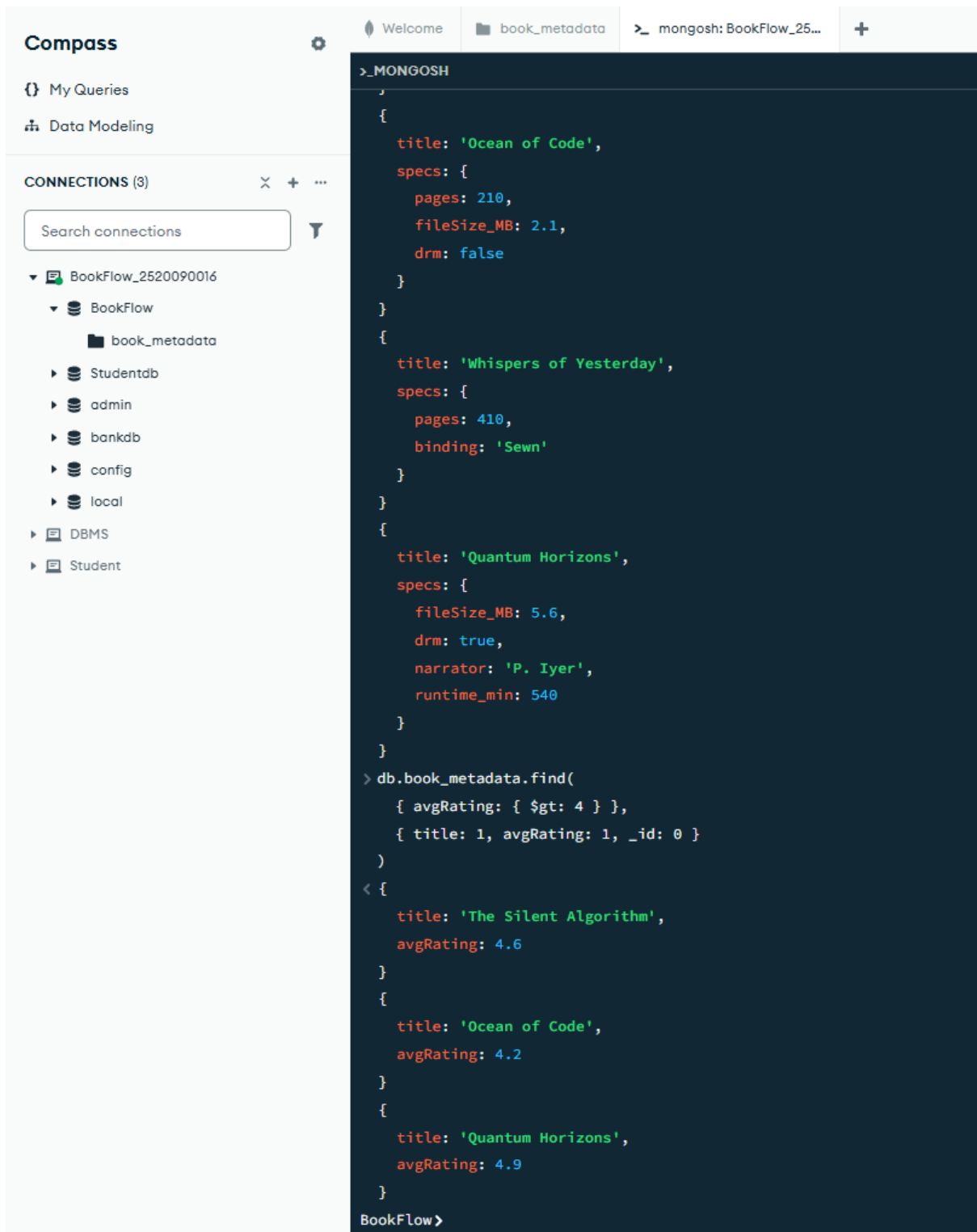
Welcomebook\_metadata>\_ mongosh: BookFlow\_25...

```
>_MONGOSH
> db.book_metadata.find(
  {},
  {
    _id: 0,
    title: 1,
    specs: 1
  }
).pretty()
< {
  title: 'The Silent Algorithm',
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  }
}
{
  title: 'Ocean of Code',
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  }
}
{
  title: 'Whispers of Yesterday',
  specs: {
    pages: 410,
    binding: 'Sewn'
  }
}
{
  title: 'Quantum Horizons',
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  }
}
```



## Task 2 – Advanced Querying

Books rated above 4 using \$gt



Books available as Digital or Audiobook using \$in

Data Modeling

CONNECTIONS (3)

Search connections

▼ BookFlow\_2520090016

▼ BookFlow

book\_metadata

▶ Studentdb

▶ admin

▶ bankdb

▶ config

▶ local

▶ DBMS

▶ Student

```
{ title: 1, avgRating: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
}
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
}
BookFlow>
```

Books rated above 4 AND available digitally

## Compass

{ } My Queries

🔗 Data Modeling

CONNECTIONS (8)

Search connections

▼ BookFlow\_2520090016

▼ BookFlow

book\_metadata

▶ Studentdb

▶ admin

▶ bankadb

▶ config

▶ local

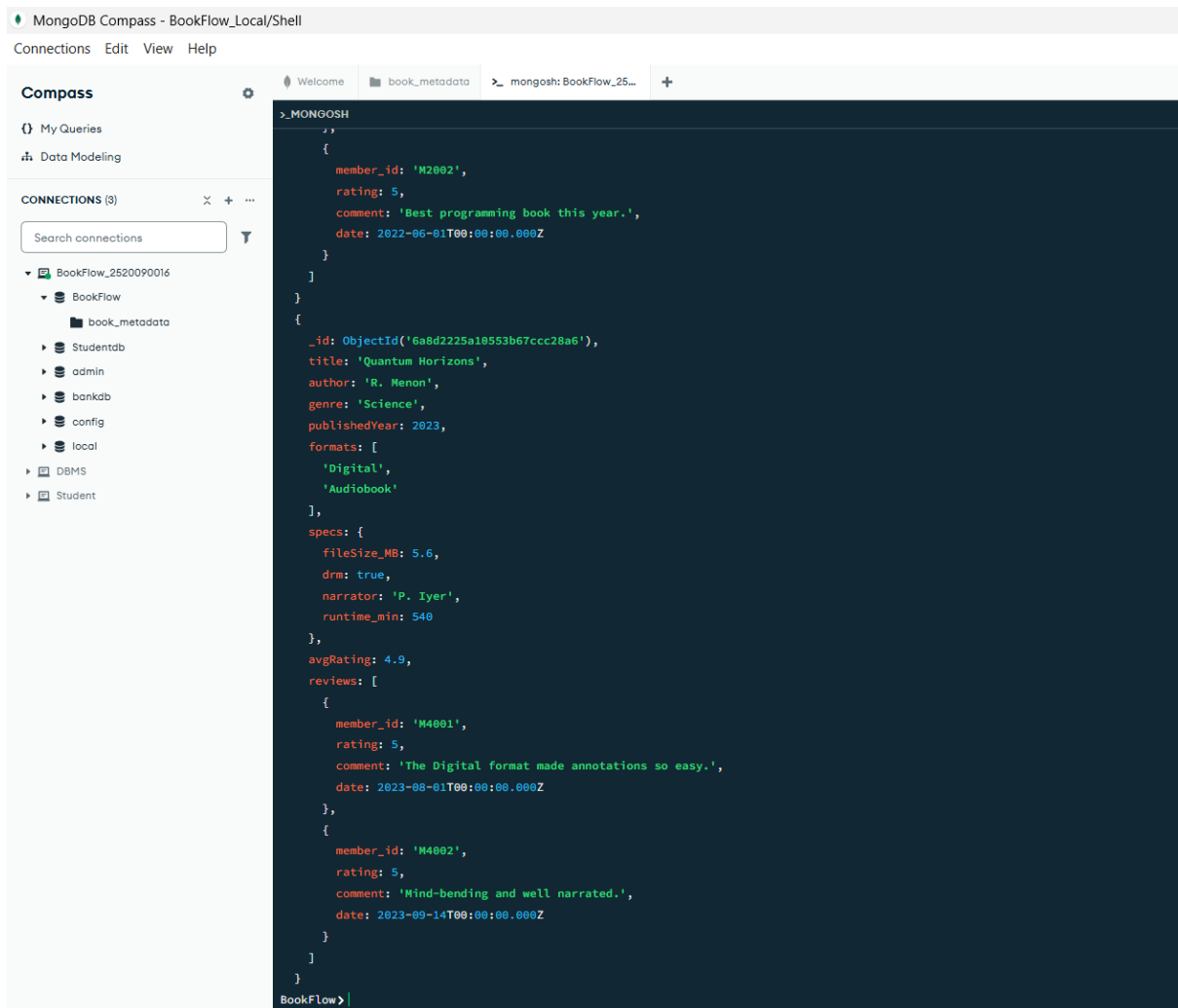
▶ DBMS

▶ Student

Welcome book\_metadata mongosh: BookFlow\_25...

&gt; MONGOSH

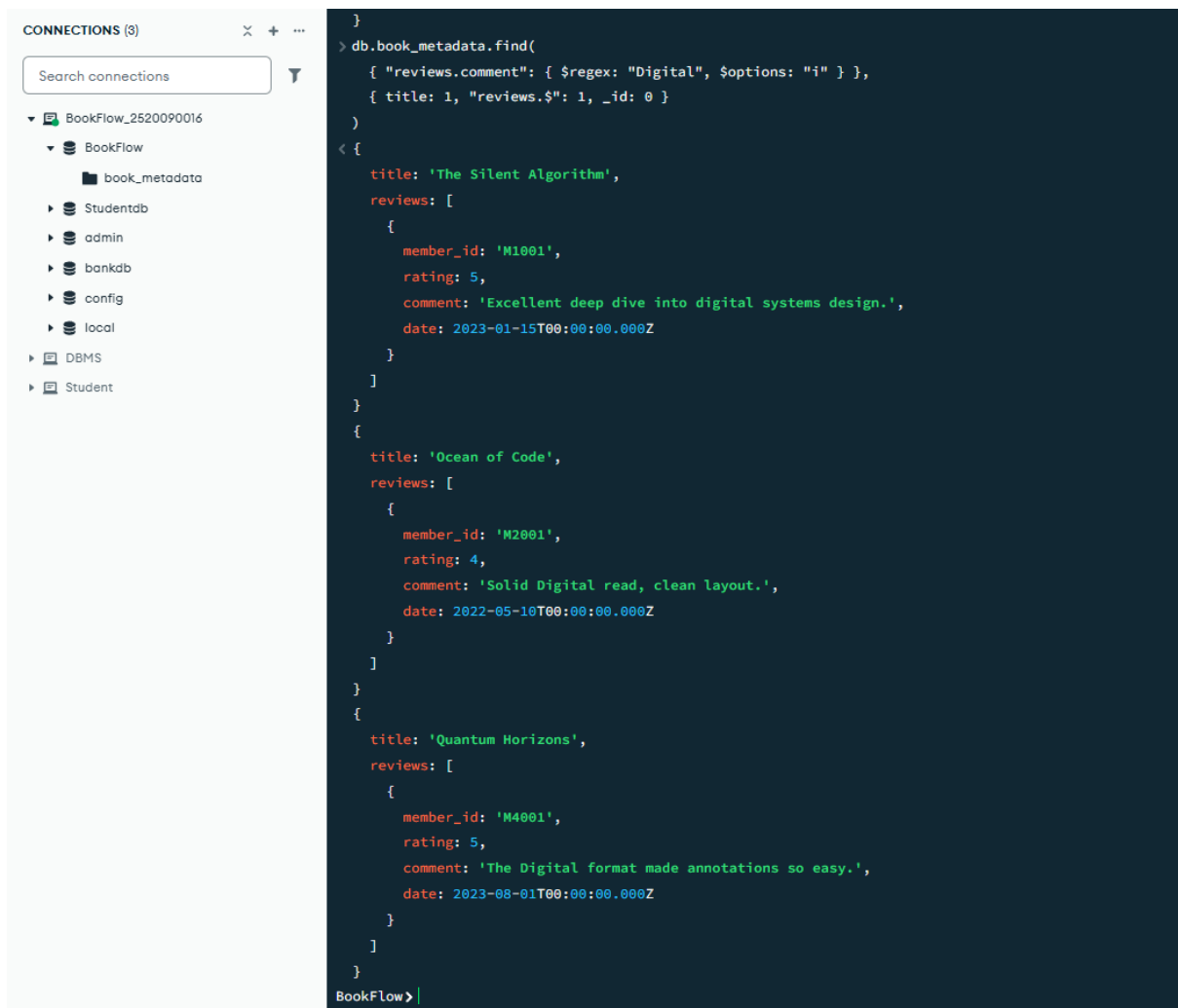
```
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d2225a10553b67ccc28a3'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8d2225a10553b67ccc28a4'),
  title: 'Ocean of Code',
```



Regex – Find books with Digital format

```
> db.book_metadata.find(
  { formats: { $regex: "Digital", $options: "i" } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
BookFlow>
```

Regex – Find reviews mentioning Digital



### Task 3 – Aggregation Pipeline

## Compass

{} My Queries

🔗 Data Modeling

## CONNECTIONS (3)

▼ BookFlow\_2520090016

▼ BookFlow

book\_metadata

▶ Studentdb

▶ admin

▶ bankdb

▶ config

▶ local

▶ DBMS

▶ Student

&gt; \_MONGOSH

```
}
> db.book_metadata.aggregate([
  {
    $match: {
      publishedYear: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  {
    $sort: {
      avgReviewRating: -1
    }
  },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: {
        $round: ["$avgReviewRating", 2]
      },
      totalReviews: 1
    }
  }
])
< {
  totalReviews: 2,
  author: "B. Menard",
```

The screenshot displays a MongoDB management tool interface. On the left, a sidebar titled 'CONNECTIONS (3)' contains a search bar and a tree view of database connections. The tree shows a connection named 'BookFlow\_2520090016' expanded, revealing a database named 'BookFlow' with a collection named 'book\_metadata'. Other connections listed include 'Studentdb', 'admin', 'bankdb', 'config', 'local', 'DBMS', and 'Student'. The main editor on the right shows a JSON query for the 'book\_metadata' collection. The query uses the '\$sort' operator to sort by 'avgReviewRating' in descending order. It also uses the '\$project' operator to select specific fields: '\_id', 'title', 'author', 'avgReviewRating', and 'totalReviews'. The 'avgReviewRating' field is further processed with the '\$round' operator to round it to 2 decimal places. The query is structured as follows:

```
{
  $sort: {
    avgReviewRating: -1
  },
},
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: {
      $round: ["$avgReviewRating", 2]
    },
    totalReviews: 1
  }
}
]>
```

Below the main query, there is a section for aggregation results, showing three documents with their respective fields and values:

```
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
BookFlow>
```

## Task 4 – Indexing

Create text index on review comments

```
> db.book_metadata.createIndex({
  "reviews.comment": "text"
})
< reviews.comment_text
> db.book_metadata.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  }
]
BookFlow> |
```

## Search using the text index

CONNECTIONS (9)

Search connections

▼ BookFlow\_2520090016

▼ BookFlow

book\_metadata

▶ Studentdb

▶ admin

▶ bankdb

▶ config

▶ local

▶ DBMS

▶ Student

```
{ v: 2, key: { _id: 1 }, name: '_id_' },
{
  v: 2,
  key: { _fts: 'text', _ftsx: 1 },
  name: 'reviews.comment_text',
  weights: { 'reviews.comment': 1 },
  default_language: 'english',
  language_override: 'language',
  textIndexVersion: 3
}
]
> db.book_metadata.find(
  { $text: { $search: "Digital" } },
  {
    title: 1,
    score: { $meta: "textScore" },
    _id: 0
  }
).sort({
  score: { $meta: "textScore" }
})
< {
  title: 'The Silent Algorithm',
  score: 1.1833333333333333
}
{
  title: 'Quantum Horizons',
  score: 0.6
}
{
  title: 'Ocean of Code',
  score: 0.6
}
}
BookFlow>
```

Verify index usage using `.explain()`

## Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

book\_metadata

Studentdb

admin

bankdb

config

local

DBMS

Student

Welcome

book\_metadata

mongosh: BookFlow\_25...

+

&gt; MONGOSH

```
> db.book_metadata.find(
  { $text: { $search: "Digital" } }
).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [
          'digit'
        ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
    },
    textIndexVersion: 3,
    inputStage: {
      stage: 'FETCH',
```

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

- BookFlow\_2520090016
  - BookFlow
    - book\_metadata
  - Studentdb
  - admin
  - bankdb
  - config
  - local
- DBMS
- Student

> MONGOSH

```
nss: 'BookFlow.book_metadata',
inputStage: {
  stage: 'IXSCAN',
  nss: 'BookFlow.book_metadata',
  keyPattern: {
    _fts: 'text',
    _ftsx: 1
  },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
  indexBounds: {}
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 11,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    isPartial: false,
    isSparse: false,
    isUnique: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  }
}
```

MongoDB Compass - BookFlow\_Local/Shell

Connections Edit View Help

Compass

My Queries

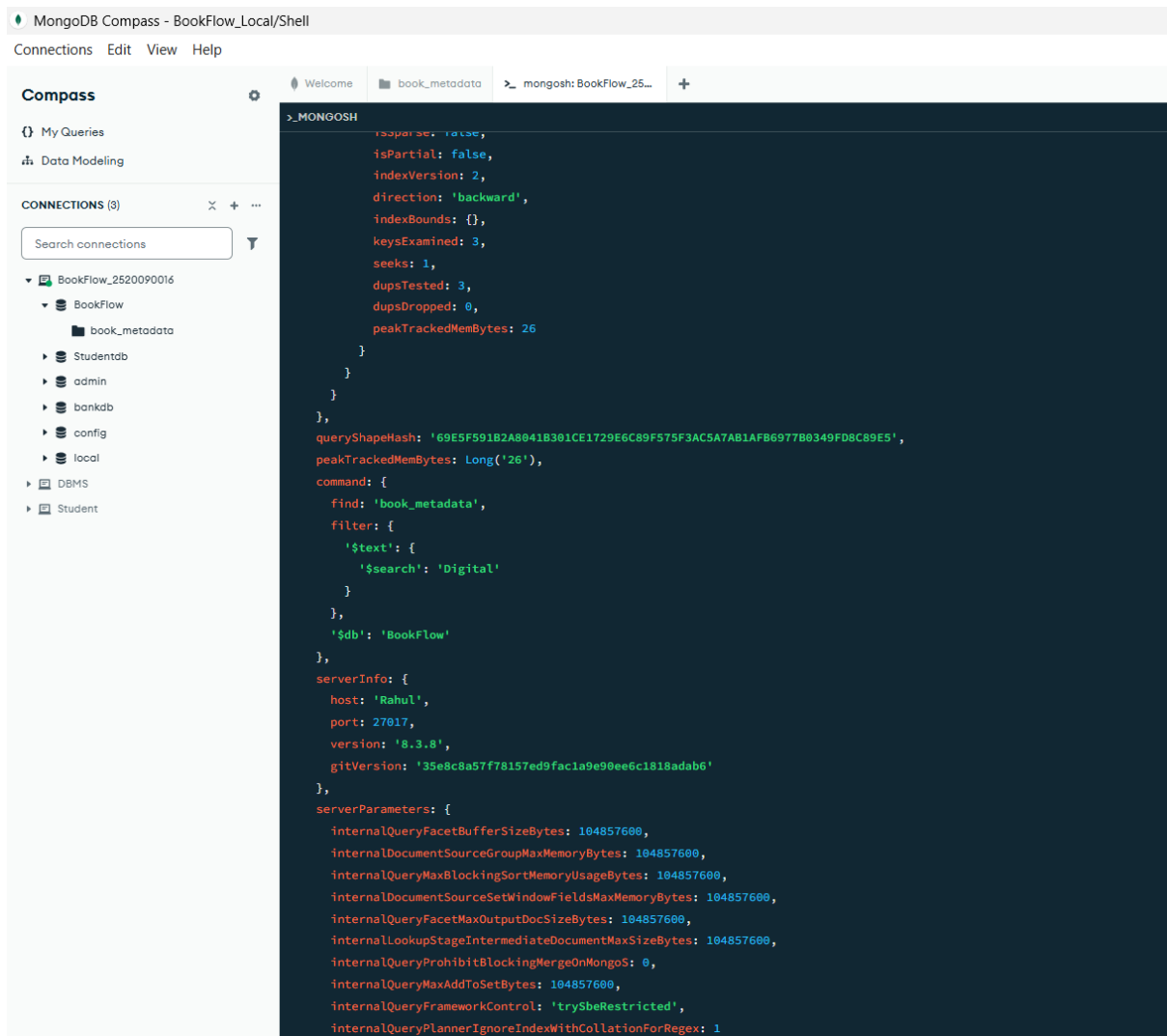
Data Modeling

CONNECTIONS (3)

Search connections

- BookFlow\_2520090016
  - BookFlow
    - book\_metadata
    - Studentsdb
    - admin
    - bankdb
    - config
    - local
  - DBMS
  - Student

```
>_MONGOOSH
{
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  docsRejected: 0,
  inputStage: {
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'BookFlow.book_metadata',
      keyPattern: {
        _fts: 'text',
        _ftx: 1
      }
    },
    indexName: 'reviews.comment_text',
```



**Compare with regex query**

## Compass

{ } My Queries

Data Modeling

## CONNECTIONS (3)

Search connections

BookFlow\_2520090016

BookFlow

Studentdb

admin

bankdb

config

local

DBMS

Student

&gt; \_MONGOSH

```
> db.book_metadata.find(
  {
    "reviews.comment": {
      $regex: "Digital",
      $options: "i"
    }
  }
).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'test.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'EOF',
      type: 'nonExistentNamespace'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 0,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 0,
    executionStages: {
      isCached: false,
      stage: 'EOF',
      nReturned: 0
    }
  }
}
```

MongoDB Compass - BookFlow\_2520090016/Shell

Connections Edit View Help

Compass

{ } My Queries

🔗 Data Modeling

CONNECTIONS (3) ✕ + ...

Search connections 🔍

- BookFlow\_2520090016
  - BookFlow
  - Studentdb
  - admin
  - bankdb
  - config
  - local
- DBMS
- Student

WELCOME

mongosh: BookFlow\_25...

> MONGOSH

```
  advanced: 0,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  type: 'nonExistentNamespace'
},
queryShapeHash: '74EC647F7B909B65D92898676CE1BC7A0BA707C0DEC11D073FDF365ED15B1A9D',
command: {
  find: 'book_metadata',
  filter: {
    'reviews.comment': BSONRegExp('Digital', 'i')
  },
  '$db': 'test'
},
serverInfo: {
  host: 'Rahul',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
test>
```